Junyoung Park

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Research Interests

I am broadly interested in data analysis that involves unique geometric structures necessitating rigorous treatment. My recent research focuses on distributional data analysis with applications to wearable device data. In addition, I have been working on compositional data and their dimension reduction methods for microbiome data analysis.

Professional Positions

Postdoctoral Research Fellow — 09/2024 –
Department of Biostatistics, University of Michigan, MI, USA
Supervisor: Irina Gaynanova

BK21 Postdoctoral Research Fellow — 03/2024 – 08/2024
Natural Science Research Institute, KAIST, Daejeon, Korea
Supervisor: Cheolwoo Park

Education

Ph.D. in Mathematical Sciences — 03/2018 – 02/2024
KAIST, Daejeon, Korea
Thesis: "Kernel Methods for Compositional Data and Dimensionality Reduction"
(Co)Advisors: Cheolwoo Park, Jeongyoun Ahn

B.S. in Mathematics — 03/2013 – 02/2018
Korea University, Seoul, Korea

Experience

Research Assistant in Statistical Learning — KAIST
Under the supervision of Prof. Cheolwoo Park and Jeongyoun Ahn — 08/2021 – 02/2024
- Kernel methods, compositional data, dimension reduction

Technical Research Personnel — KAIST
For military duty in South Korea — 03/2020 – 02/2023

Research Assistant in Algebraic Geometry — KAIST
Under the supervision of Prof. Sijong Kwak — 09/2018 – 07/2021
- Projective algebraic geometry, syzygies, applied algebraic geometry

University Financial Engineering Association (U.FE.A) — Seoul, Korea
Team leader (Master's-level financial engineering & mathematics studies) — 03/2016 – 08/2017

Awards and Scholarships

Competitive Travel/Paper Awards

- **IMS New Researchers Travel Award** — 2026
- **IMSI Long Program**, visit to Connectomics — 2026
- **IMS International New Researchers Conference** — 2025
- **Best Ph.D. Dissertation Award**, Korean Statistical Society; prize: 3,000,000 (KRW) — 2025

Other Awards/Scholarships

- **Presentation Award for Graduate Students**, Korean Statistical Society; prize: 300,000 (KRW) 2022
- **The Outstanding Teaching Assistant Award**, Calculus II, KAIST Fall, 2020
- **University Students Contest of Mathematics**, Silver Awards, Korean Mathematical Society (KMS) 2016, 2017
- **Presidential Science Undergraduate Fellowship**, fully funded for 8 semesters 2013–2018
- **The Korean Mathematical Olympiad (KMO)** 2nd round of the middle school division, Gold Awards 2009

Research Papers

Preprints/Submitted:

9. **Park, J.** and Gaynanova, I. (2026+) "Fréchet regression of multivariate distributions with nonparanormal transport." Submitted. Preprint available on arXiv:2603.07014.

8. Shao, E., **Park, J.**, Punjabi, N., Jiang, H, and Gaynanova, I. (2026+) "Fast distance computation between multivariate distributions via nonparanormal Transport." Preprint available on arXiv:2603.00322.

7. Xu, C., Schmidt, B.M., Krambrink, A., **Park, J.**, Song, P.X.-K., Jones, T.L.Z., Holmes, C., Chen, K., Ye, W., Kolenic, G., Roy, S., Sen, C., Pop-Busui, R., Spino, C., and Gaynanova, I. (2026+) "Profiling Longitudinal Wound Healing: Analysis of Healing Rates and the Predictive Power of Wound Area and Duration in the Diabetic Foot Consortium." Submitted.

6. **Park, J.**, Park, C., and Ahn, J. (2026+) "A geometry-preserving framework for sufficient dimension reduction of compositional data." Submitted. Preprint available on arXiv:2509.05563.

5. **Park, J.**, Kok, N., and Gaynanova, I. (2026+) "Beyond fixed thresholds: optimizing summaries of wearable device data via piecewise linearization of quantile functions." Submitted. Preprint available on arXiv:2501.11777.

Peer-reviewed publications:

4. Yeon, K., Gao, F., Kim, R., Lee, D., **Park, J.**, Brown, D.A., and Park, C. (2026+) "A Bayesian Approach to Semi-Supervised Domain Adaptation in Streaming Data." *Journal of Statistical Computation and Simulation*, Accepted.

3. **Park, J.**, Ahn, J., and Park, C. (2023), "Kernel Sufficient Dimension Reduction and Variable Selection for Compositional Data via Amalgamation." *International Conference on Machine Learning* (ICML), pp. 27034-27047, PMLR
   Link: https://proceedings.mlr.press/v202/park23a.html.

2. Kang, I., Choi, H., Yoon, Y.-J., **Park, J.**, Kwon, S.-S., and Park, C. (2023), "Frechet Distance-Based Cluster Analysis for Multi-Dimensional Functional Data." *Statistics and Computing*, 33(4), 75.
   Link: https://doi.org/10.1007/s11222-023-10237-z

1. **Park, J.**, Yoon, C., Park, C., and Ahn, J. (2022), "Kernel Methods for Radial Transformed Compositional Data with Many Zeros." *International Conference on Machine Learning* (ICML), pp. 17458-17472, PMLR
   Link: https://proceedings.mlr.press/v162/park22d.html.

Talks

Invited talks/seminars:

- **1st IMS International New Researchers Conference** 12/2025
  - Title: Fréchet regression of multivariate distributions under the semiparametric nonparanormal model
- **KAIST, Graduate School of Data Science** 07/2025
  - Title: Distributional data approaches for wearable device data: optimal thresholds and beyond

Contributed talks/posters:

- **Mobile Technologies Research Innovation Collaborative (MeTRIC) Symposium** (poster) 01/2026
  - Title: Beyond fixed thresholds: optimizing summaries of wearable device data

- **2025 IMS International Conference on Statistics and Data Science (ICSDS)** 12/2025
  - Title: Fast Fréchet regression of multivariate distributions with application to wearable device data
- **2025 Joint Statistical Meetings** 08/2025
  - Title: Beyond fixed thresholds: optimizing summaries of wearable device data
- **2024 Joint Statistical Meetings** 08/2024
  - Title: Interpretable dimension reduction for compositional data
- **2023 Winter Conference, the Korean Statistical Society** 12/2023
  - Title: Interpretable composition-to-composition dimension reduction via conditional covariance operator
- **40th International Conference on Machine Learning (ICML)** (poster) 07/2023
  - Title: Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **2023 Summer Conference, the Korean Statistical Society** 06/2023
  - Title: Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **KAIST Mathematical Sciences Graduate student Seminar** 10/2022
  - Title: Kernel methods for radial transformed compositional data with many zeros
- **39th International Conference on Machine Learning (ICML)** (spolight talk) 06/2022
  - Title: Kernel methods for radial transformed compositional data with many zeros
  - Presented also at 2022 Summer Conference, the Korean Statistical Society, (**presentation award**)

Teaching

- **Teaching Assistant** at KAIST (selected list)
  - Statistical Data Science Practice (DS516) Spring 2023
  - Probability and Statistics (MAS250) Fall 2021
  - Abstract Algebra I (MAS311) Spring 2021
  - Mathematical Statistics (MAS355) Fall 2019
  - Matrix Group Theory (MAS435) Spring 2019
    - Gave a guest lecture on connectedness of Lie groups (in English)
  - Abstract Algebra II (MAS312) Fall 2018
    - Gave several guest lectures throughout the semester

Academic Services

- **Journal Refereeing**
  - Journal of Royal Statistical Society, Series B (1); Annals of Applied Statistics (2); Biometrics (3); WIREs Computational Statistics (1)

Computing Skills

- Python (for machine learning, statistics), R (for statistics), previous experiences with Matlab and C
- Optimization acceleration: with deep learning frameworks (`TensorFlow` and `Pytorch`), with just-in-time compiler (`numba`)
- Linux environment and high-performance computing (HPC) cluster systems

Languages

- English
- Korean

Hobbies

- Singing, better with playing guitar.
- Running, hiking, and climbing.